

INTERNSHIP

CUSTOMER CHURN & REVENUE ANALYTICS

An End-to-End Data Analytics Project

Field	Details
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Executive Summary

This project analyzes customer churn and revenue-related patterns using a structured data analytics workflow. The work covers data acquisition, data-quality assessment, cleaning, exploratory analysis, SQL analysis, visualization, and dashboard development.

The final report is intentionally designed so that every important business statement can be traced to a dataset field, an analytical method, a query/calculation, and a visual or tabular output. Findings should be stated as evidence from the analyzed population rather than as unsupported claims of causation.

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1. Introduction

Organizations collect customer information through subscriptions, services, billing, and support interactions.

The purpose of this project is to transform customer-level data into reliable descriptive and diagnostic insights

about churn and revenue. The analysis follows an end-to-end workflow so that the final conclusions can be reproduced

from the underlying data and code.

2. Business Context and Problem Statement

Customer churn can affect recurring revenue and increase the need for customer acquisition. The analytical problem

is to understand the historical characteristics associated with churn and to quantify how customer groups differ in

revenue-related measures.

Primary business question:

How can customer data be analyzed to understand churn patterns and revenue distribution while maintaining a

traceable, evidence-based analytical process?

3. Objectives and Analytical Questions

Objectives:

- Assess the quality and structure of the customer dataset.
- Prepare analysis-ready data using documented cleaning rules.
- Measure customer and churn KPIs.
- Compare churn across relevant customer segments.
- Analyze revenue-related measures.
- Perform SQL-based validation and aggregation.
- Build a dashboard for repeatable reporting.
- Document findings, limitations, and recommendations.

Analytical questions:

- What is the observed churn rate in the analysis population?
- How does churn vary by contract type?
- How does churn vary across tenure groups?
- How do monthly and total charges differ by churn status?
- Which customer segments contribute the largest recorded revenue?
- Which observed relationships deserve further investigation?

4. Scope and Assumptions

In scope:

Customer-level descriptive and diagnostic analysis, data preparation, SQL queries, visualization, and dashboarding.

Out of scope:

The project does not claim that an observed variable causes churn. It also does not claim to predict future churn

unless a separately validated predictive model is developed.

Assumptions must be recorded explicitly, including the observation period, population definition, currency/unit of financial measures, treatment of missing values, and any exclusions.

5. Dataset and Data Provenance

Record the exact source of the dataset used in the final analysis. If a public dataset is used, record the dataset

name, source URL, download date, version if available, and license/usage conditions. If the organization supplied the

data, record the approved internal source and access date without exposing confidential information.

Never change the source data directly. Preserve an immutable raw copy and perform transformations into a separate processed dataset.

5.1 Dataset Control Record

Item	Value to Complete
Dataset name	[Exact dataset name]
Source / provider	[URL or organization]
Access / download date	[Date]
Raw file name	[File name]
Raw row count	[Measured value]
Raw column count	[Measured value]
Analysis row count	[Measured value]
Analysis period	[If applicable]
Currency / units	[e.g., INR / USD / not applicable]
License / permission	[Record terms]

6. Data Dictionary

Complete this table from the actual dataset. Do not infer definitions that are not supported by the source.

Column	Data Type	Business Meaning	Allowed / Typical Values	Quality Notes
CustomerID	[text]	Unique customer identifier	[values]	[check uniqueness]
Gender	[category]	Customer gender field	[categories]	[missing/consistency]
Tenure	[numeric]	Customer tenure	[unit]	[range check]

Contract	[category]	Contract type	[categories]	[category validation]
PaymentMethod	[category]	Recorded payment method	[categories]	[consistency]
MonthlyCharges	[numeric]	Recorded monthly charge	[currency]	[range/outliers]
TotalCharges	[numeric]	Recorded total charges	[currency]	[missing/type]
Churn	[category]	Observed churn status	[e.g. Yes/No]	[class balance]

7. Data Quality Assessment

Before analysis, calculate and record missingness, duplicate records, invalid data types, impossible ranges, category inconsistencies, and potential outliers. Every cleaning decision should have a reason and a measurable effect.

Quality Check	Method	Before	Action	After	Evidence / Output
Missing values	isna().sum()	[measured]	[rule]	[measured]	[table/figure]
Duplicate rows	duplicated().sum()	[measured]	[rule]	[measured]	[table]
Data types	info()/schema	[measured]	[conversion]	[measured]	[log]
Numeric ranges	min/max/quantiles	[measured]	[rule]	[measured]	[table]
Category consistency	unique/value_counts	[measured]	[rule]	[measured]	[table]

8. Data Preparation and Transformation

Document transformations in the same order they are executed in code. Typical steps include type conversion, missing-value treatment, duplicate handling, category standardization, and creation of derived fields such as tenure groups. Each transformation should be reproducible from the raw data.

Example transformation log:

Step	Transformation	Reason	Impact	Script / Notebook
1	Convert TotalCharges to numeric	Enable numeric analysis	[measured]	src/data_cleaning.py
2	Handle missing values	Avoid invalid calculations	[measured]	src/data_cleaning.py
3	Create TenureGroup	Support segment analysis	[measured]	src/data_cleaning.py

9. Exploratory Data Analysis

EDA should move from descriptive statistics to comparisons and relationships. Each chart must have a clear question, a defined population, labeled units, and a short interpretation. Avoid decorative charts that do not answer an analytical question.

9.1 Overall Customer and Churn Profile

Measure total customers, churned customers, active customers, and observed churn rate.

Evidence to insert: [Figure/Table ID] | Population: [definition] | Result: [measured value] | Interpretation: [evidence-based statement]

9.2 Churn by Contract

Compare churn counts and percentages within contract groups.

Evidence to insert: [Figure/Table ID] | Population: [definition] | Result: [measured value] | Interpretation: [evidence-based statement]

9.3 Churn by Tenure

Compare churn across documented tenure groups.

Evidence to insert: [Figure/Table ID] | Population: [definition] | Result: [measured value] | Interpretation: [evidence-based statement]

9.4 Charges by Churn Status

Compare monthly and total charge distributions.

Evidence to insert: [Figure/Table ID] | Population: [definition] | Result: [measured value] | Interpretation: [evidence-based statement]

9.5 Payment Method and Churn

Compare observed churn proportions by payment method.

Evidence to insert: [Figure/Table ID] | Population: [definition] | Result: [measured value] | Interpretation: [evidence-based statement]

9.6 Service Characteristics

Assess relationships between selected services and churn.

Evidence to insert: [Figure/Table ID] | Population: [definition] | Result: [measured value] | Interpretation: [evidence-based statement]

10. SQL Analysis

SQL queries should be stored separately in the repository and the report should show only the important queries and their validated outputs. The database table definition and import process should also be documented.

Customer count

```
SELECT COUNT(*) AS total_customers FROM customers;
```

Churned customers

```
SELECT COUNT(*) AS churned_customers FROM customers WHERE Churn = 'Yes';
```

Churn by contract

```
SELECT Contract, COUNT(*) AS customers, SUM(CASE WHEN Churn='Yes' THEN 1 ELSE 0 END) AS churned_customers FROM customers GROUP BY Contract;
```

Revenue by contract

```
SELECT Contract, SUM(TotalCharges) AS total_revenue FROM customers GROUP BY Contract;
```

11. Dashboard and KPI Design

The dashboard should communicate the same definitions used in Python and SQL. KPI definitions must be written down so that the number shown in Power BI can be reconciled with the analytical dataset.

KPI	Definition	Calculation	Source	Validation
Total Customers	Count of analysis records/customers	COUNT(CustomerID)	[table]	[match Python/SQL]
Churn Rate	Churned customers / eligible customers	Churned ÷ eligible × 100	[table]	[match]
Total Revenue	Sum of recorded total charges	SUM(TotalCharges)	[table]	[match]
Average Monthly Charge	Mean monthly charge	AVG(MonthlyCharges)	[table]	[match]
Average Tenure	Mean customer tenure	AVG(Tenure)	[table]	[match]

12. Findings and Evidence

This section must contain the actual findings after the analysis is run. Do not fill this section with expected or assumed numbers. For every important finding, record the metric, comparison group, time/population definition, and evidence source.

Finding ID	Question	Observed Result	Evidence	Interpretation	Caution
F01	[Question]	[Measured result]	[Figure/Table]	[What the data supports]	[Limitations]
F02	[Question]	[Measured result]	[Figure/Table]	[What the data supports]	[Limitations]
F03	[Question]	[Measured result]	[Figure/Table]	[What the data supports]	[Limitations]

13. Business Interpretation

Translate statistical/descriptive findings into business language without overstating causality. Use phrases such as 'the analysis shows an observed difference' or 'this segment has a higher observed churn proportion' when that is what the data supports. Avoid statements such as 'X causes churn' unless a causal design supports that claim.

14. Recommendations

Recommendations should follow directly from validated findings. Each recommendation should state the evidence it is based on, the business action proposed, the intended metric to monitor, and any assumptions.

Recommendation	Evidence	Proposed Action	Success Metric	Owner / Stakeholder
R01	[Finding ID]	[Action]	[Metric]	[Role]
R02	[Finding ID]	[Action]	[Metric]	[Role]
R03	[Finding ID]	[Action]	[Metric]	[Role]

15. Validation and Reproducibility

A responsible analytics project should be reproducible. The final repository should contain the raw-data access instructions (not confidential raw data), processing code, SQL scripts, notebooks, requirements file, dashboard, and report. Results should be cross-checked between Python, SQL, and the dashboard where applicable.

Validation	Expected Check	Status
Row-count reconciliation	Raw → cleaned → analysis counts documented	[Pending/Complete]
KPI reconciliation	Python, SQL, dashboard agree within documented rules	[Pending/Complete]
Missing-value reconciliation	Cleaning impact recorded	[Pending/Complete]
Duplicate reconciliation	Duplicate treatment documented	[Pending/Complete]
Chart verification	Chart values match source table	[Pending/Complete]
Code execution	Clean run from documented environment	[Pending/Complete]

16. Limitations and Risks

Document dataset coverage, sample size, missingness, measurement limitations, possible selection bias, time period, privacy considerations, and the difference between correlation and causation. Historical churn patterns should not automatically be treated as future predictions.

17. Future Scope

Potential extensions include a separately validated churn-prediction model, customer segmentation, automated data refresh, cloud deployment, and monitoring. Any predictive model should use an appropriate train/test strategy, avoid leakage, report suitable evaluation metrics, and be validated before being used for business decisions.

18. Conclusion

The project demonstrates an end-to-end data analytics workflow for customer churn and revenue analysis. Its value depends on the quality of the data, transparency of transformations, correctness of calculations, and disciplined interpretation of results. The completed version of this report should contain measured results generated from the final analysis rather than assumed values.

19. References

Record the exact sources used for the dataset, Python libraries, SQL database documentation, Power BI documentation, and any methodological references. Include access dates where appropriate.

Appendix A — Project Repository Structure

customer-churn-analytics/

```
|— data/  
|  |— raw/
```

```
|   └─ processed/
|   └─ notebooks/
|       └─ 01_data_quality.ipynb
|       └─ 02_data_cleaning.ipynb
|       └─ 03_eda.ipynb
|       └─ 04_validation.ipynb
|   └─ src/
|       └─ data_cleaning.py
|       └─ exploratory_analysis.py
|       └─ utils.py
|   └─ sql/
|       └─ customer_analysis.sql
|   └─ dashboard/
|       └─ customer_churn_dashboard.pbix
|   └─ outputs/
|       └─ figures/
|       └─ tables/
|   └─ reports/
|       └─ internship_report.docx
|   └─ requirements.txt
|   └─ README.md
|   └─ .gitignore
```

Appendix B — Required Evidence Checklist

- ☐ Dataset source and provenance recorded
- ☐ Raw dataset preserved separately from processed data
- ☐ Data dictionary completed
- ☐ Missing-value analysis completed
- ☐ Duplicate analysis completed
- ☐ Data-type validation completed
- ☐ Cleaning rules documented
- ☐ EDA charts generated from code
- ☐ SQL queries saved and tested
- ☐ Python/SQL/dashboard KPI reconciliation completed
- ☐ Dashboard screenshots inserted
- ☐ Actual measured findings inserted

☐ Limitations documented

☐ References completed

Appendix C — Responsible Analysis Note

This report is designed as an evidence-based template. Bracketed fields must be completed using the actual project records. Numerical values, findings, screenshots, and conclusions should not be fabricated. If a result cannot be validated from the data or code, it should be marked as unavailable rather than estimated.